

I built a governed AI knowledge assistant

Useful answers — without giving every user access to every document.

IDENTITY



ACCESS



EVIDENCE



ANSWER

Access is decided before retrieval — not by the AI model.

Python · FastAPI · Entra ID · Azure App Service · Blob Storage · Bicep

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AI Knowledge Assistant

A useful assistant also needs to know when not to answer

The same question can produce three safe outcomes, depending on identity, permission and evidence.



Grounded answer

The user is authorised and the answer includes a source citation.



Access denied

Relevant information exists, but the user's trusted role cannot access it.



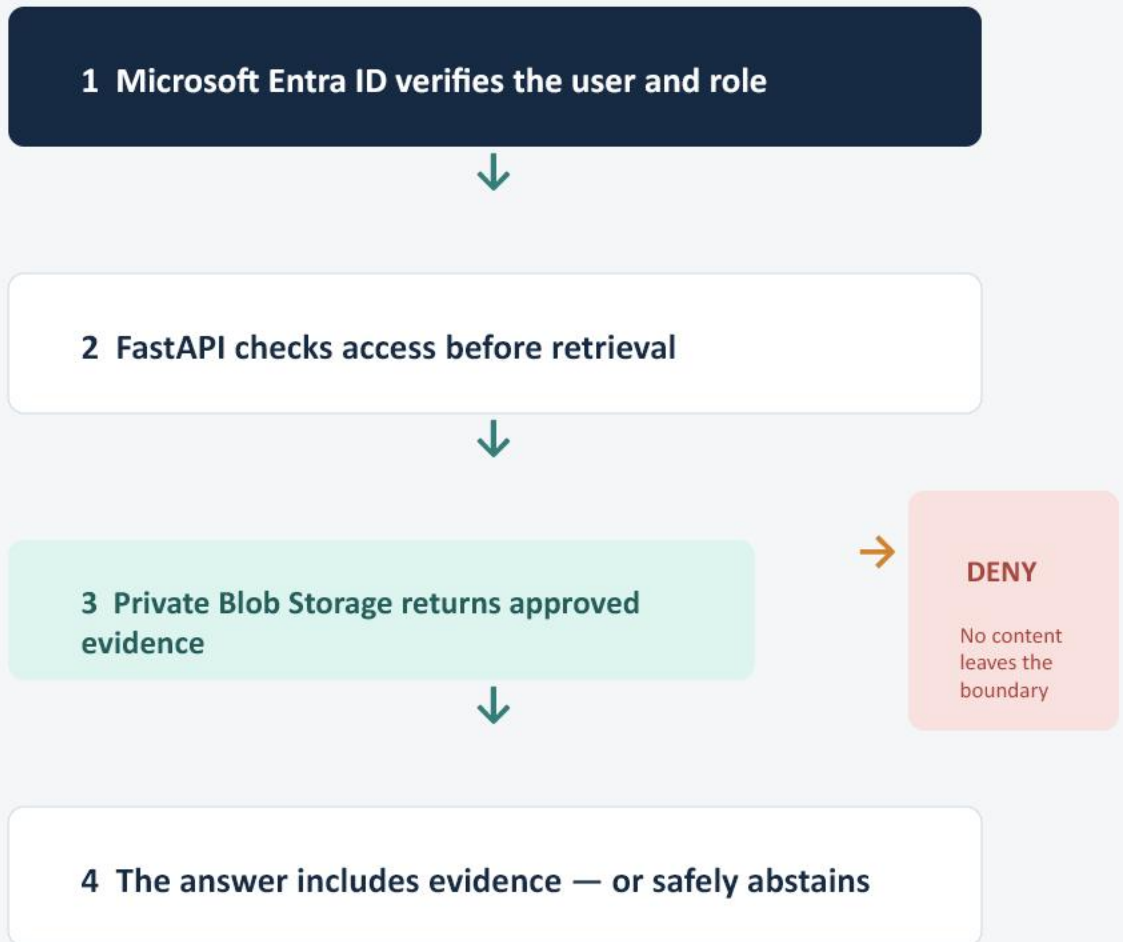
Safe uncertainty

Approved evidence is unavailable, so the assistant declines instead of guessing.

The control is simple to explain to staff — and difficult to bypass accidentally.

Identity and access control sit in front of the knowledge base

The application retrieves only the documents approved for the user's Entra role.



The governance rules are demonstrated — and tested

The project covers hierarchy, departmental access and an explicit executive exception.

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trusted roles

Employee, manager, senior executive, HR, IT and finance

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fictional documents

Public, internal, restricted and highly confidential

102

automated tests

Identity, access, API, Azure adapters and runtime safety

Least privilege still applies at the top

A senior executive can review the financial planning summary — but highly confidential payroll controls remain finance-only.

Responsible AI is an architecture choice

The model is only one part of a trustworthy system.

01 **Verify identity**

Use a trusted organisational role — never a self-selected production persona.

02 **Filter before retrieval**

Keep unauthorised documents outside the answer-generation context.

03 **Show evidence**

Citations, denial and safe refusal make the outcome understandable and auditable.

Explore the project

github.com/JanithW1996/ai-knowledge-assistant